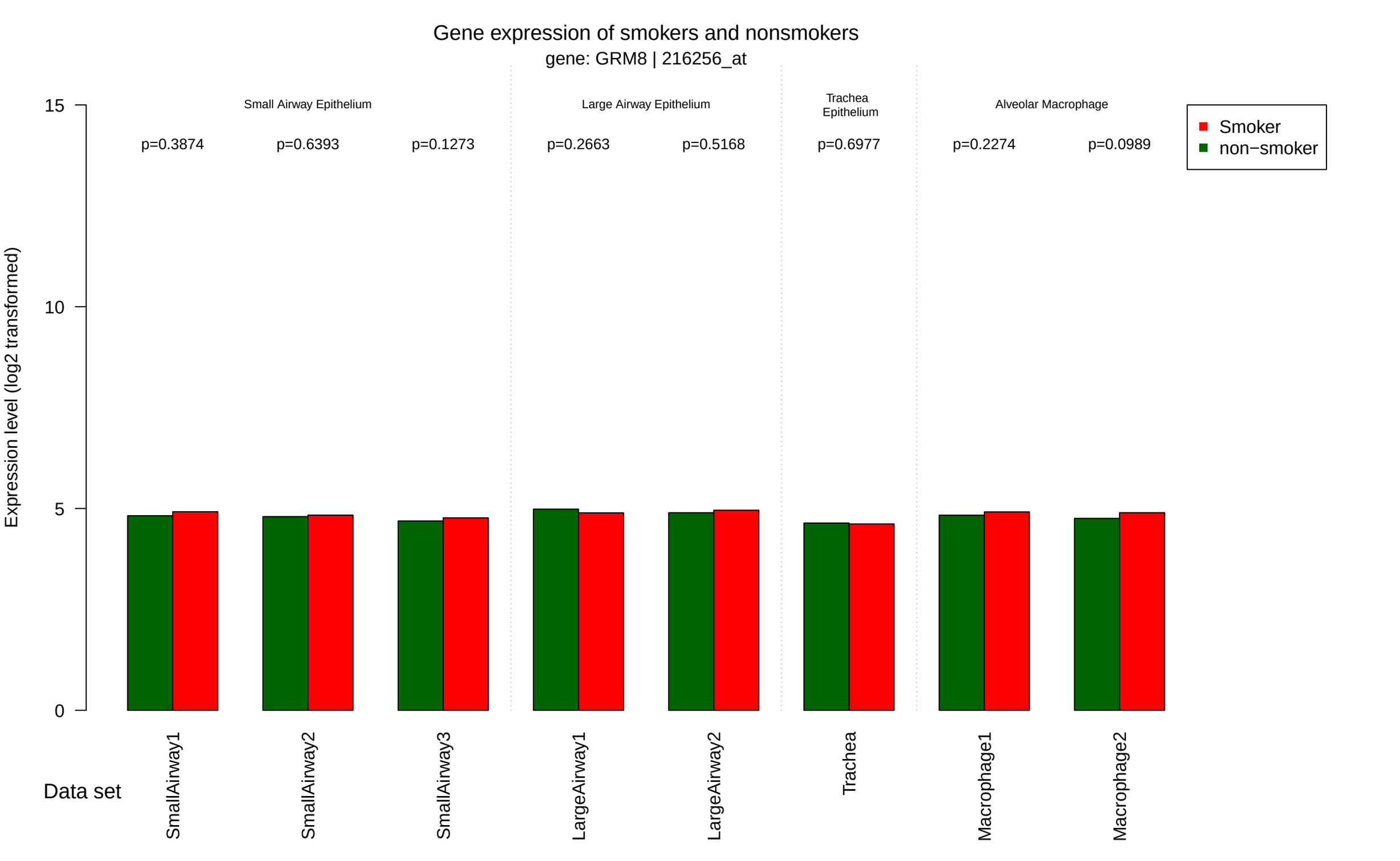
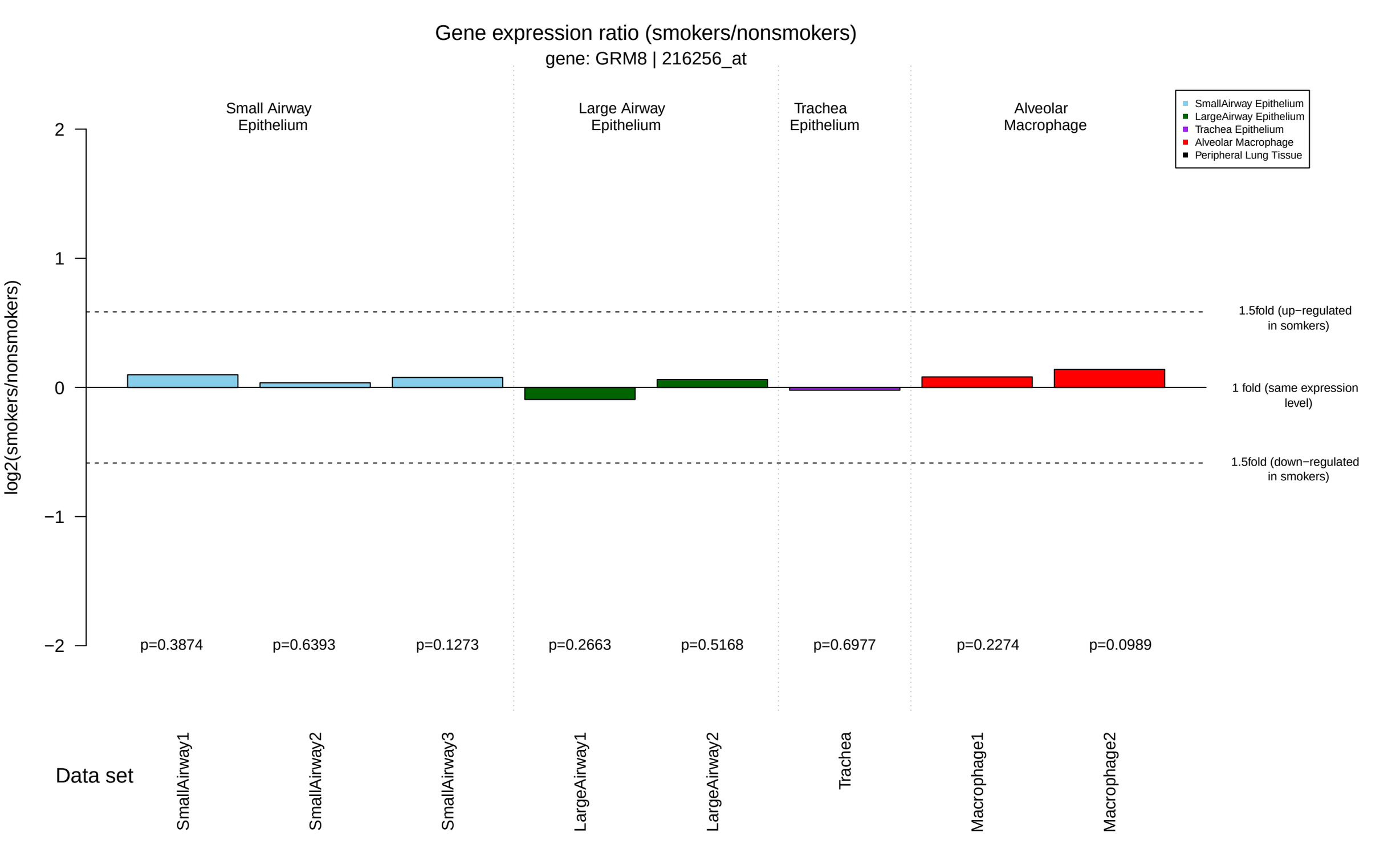
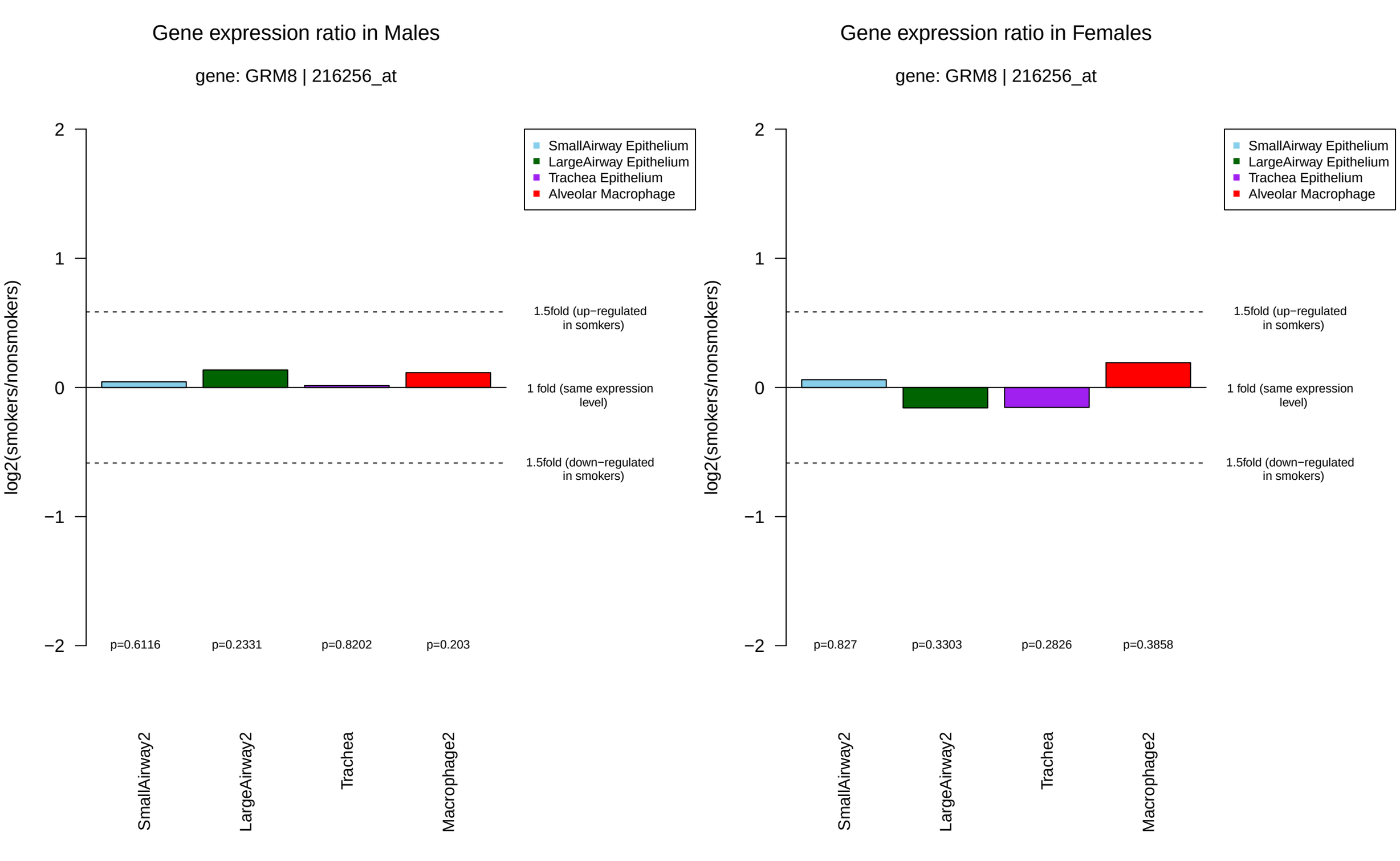
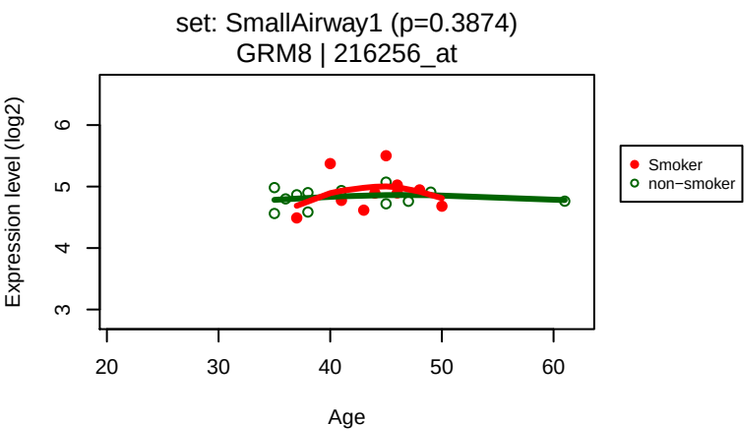
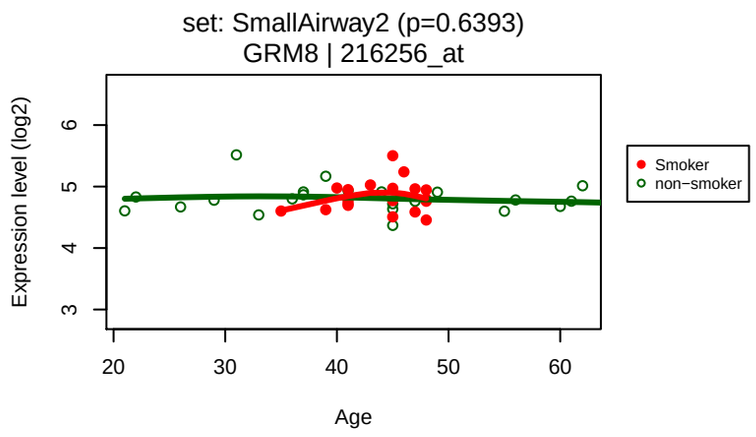
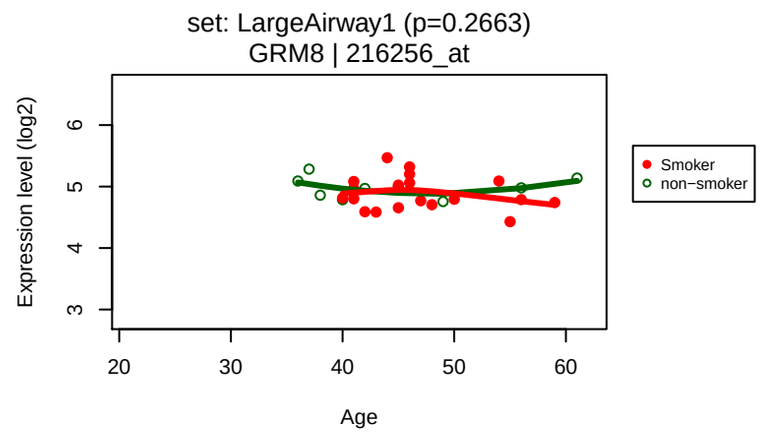

Gene expression of smokers and nonsmokers
gene: GRM8 | 216256_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3874
p=0.6393
p=0.1273
p=0.2663
p=0.5168
p=0.6977
p=0.2274
p=0.0989
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

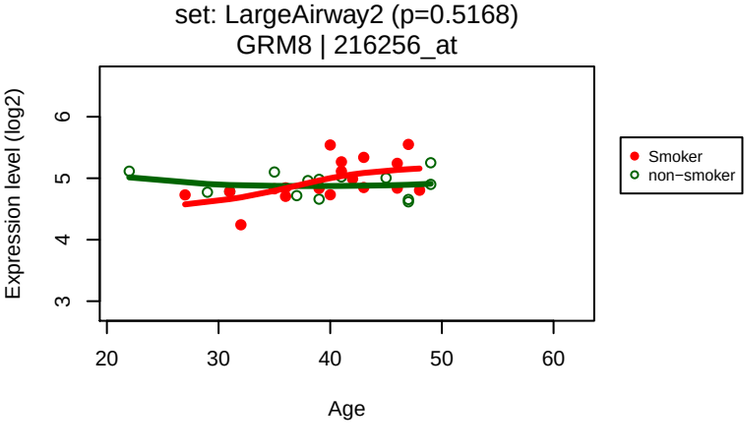

Gene expression ratio (smokers/nonsmokers)
gene: GRM8 | 216256_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3874
p=0.6393
p=0.1273
p=0.2663
p=0.5168
p=0.6977
p=0.2274
p=0.0989
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

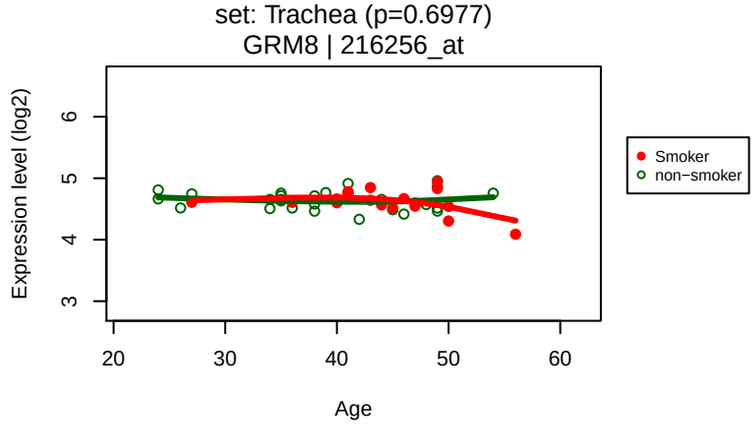

Gene expression ratio in Males
gene: GRM8 | 216256_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6116
p=0.2331
p=0.8202
p=0.203
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM8 | 216256_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.827
p=0.3303
p=0.2826
p=0.3858
SmallAirway2
LargeAirway2
Trachea
Macrophage2

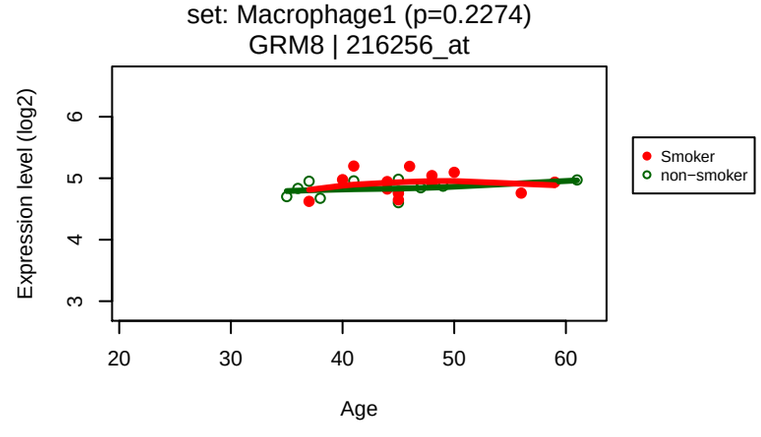

set: SmallAirway1 (p=0.3874)
GRM8 | 216256_at
Expression level (log2)
Age
Smoker
non−smoker

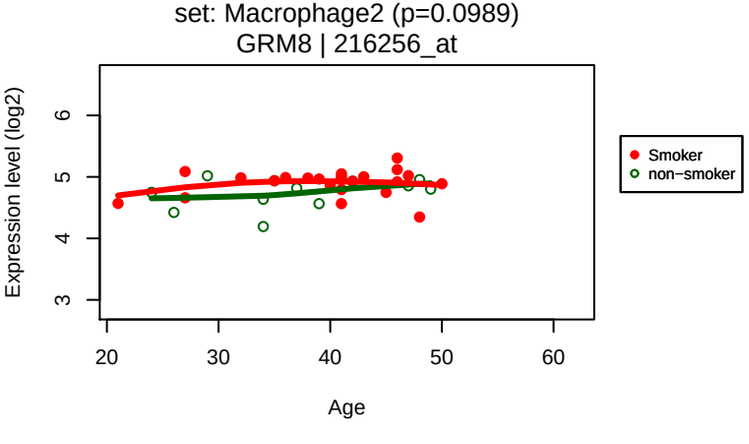

set: SmallAirway2 (p=0.6393)
GRM8 | 216256_at
Expression level (log2)
Age
Smoker
non−smoker

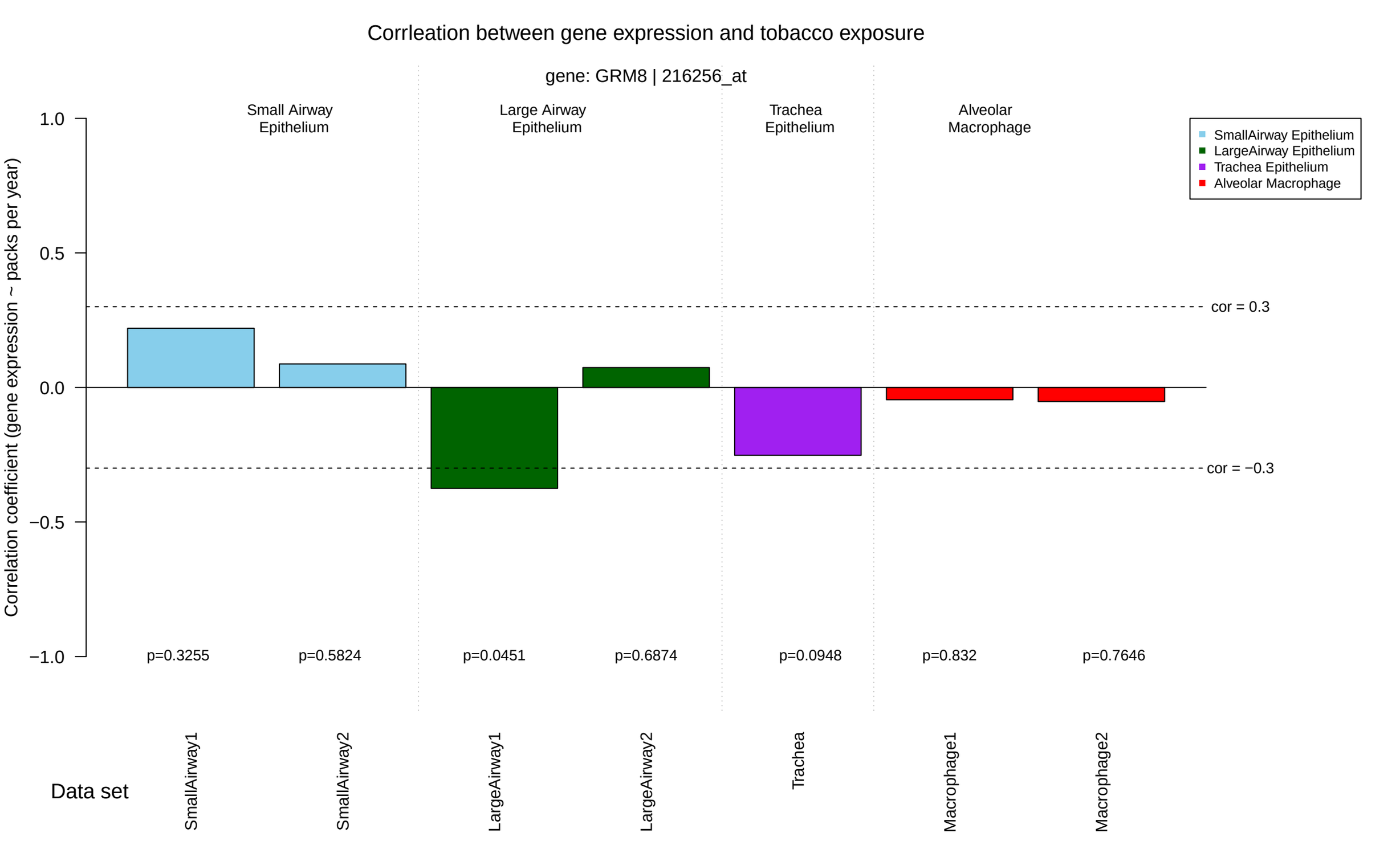

set: LargeAirway1 (p=0.2663)
GRM8 | 216256_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5168)
GRM8 | 216256_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6977)
GRM8 | 216256_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2274)
GRM8 | 216256_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0989)
GRM8 | 216256_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM8 | 216256_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3255
p=0.5824
p=0.0451
p=0.6874
p=0.0948
p=0.832
p=0.7646
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2